Political Alpha.

A test of whether U.S. Senate stock disclosures predict sector ETF behavior. And why, mostly, they don't.

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github.com/ananth-dhananjay/political-alpha

The question.

*Do **U.S. Senate stock disclosures** carry predictive signal for forward sector ETF returns, and does the 45-day disclosure delay erode that edge?*

WHY IT MATTERS

The STOCK Act (2012) requires senators to disclose trades within 45 days. The question is whether that lag is short enough to matter. If trades still carry predictive signal once public, the regime is failing. If not, it is working. This project measures which.

APPROACH

Scrape 2,284 Senate Periodic Transaction Reports. Tag each trade to a sector. Build a weekly signal and regress forward ETF returns on lagged signal, HAC standard errors.

PRE-REGISTERED EXPECTATION

With a 6-week disclosure lag, the null is no significant predictive signal. Rejection would imply leakage or skill.

Four sources.

Weekly panel, 2023–2026. One of the four (Congress.gov) turned out to be unusable (see §06).

SOURCE	ACCESS	COVERAGE	ROLE
Yahoo Finance	yfinance library	1,507 days × 7 tickers	ETFs, VIX, SPY, ^IRX
Senate EFD	HTTP + CSRF session	2,284 trades · 34 senators · $207M notional	Trade signal
Congress.gov	REST API with key	*5,000 bills pulled, 276 tagged (5.5%)*	*Legislative signal (unused (see §06))*
Ken French Library	public ZIP	1,509 daily rows: Mkt-RF / SMB / HML / RF	Factor controls

Five sectors tracked: **Defense** (ITA) · **Finance** (XLF) · **Energy** (XLE) · **Healthcare** (XLV) · **Tech** (XLK)

How the pipeline runs.

Four stages. Every step reproducible from one command: python main.py

01 FETCH

Pull raw data

yfinance: daily closes for ITA, XLF, XLE, XLV, XLK, VIX, SPY, ^IRX. Senate EFD scraper: GET /search/ for CSRF token, POST terms-agreement, paginated POST /report/data/, per-PTR HTML table parse. Congress.gov REST pull. Ken French daily factor ZIP.

02 PARSE & TAG

Structure + label

Normalize each PTR row: senator, ticker, direction (Purchase / Sale-Full / Sale-Partial), date, amount bucket → midpoint ($1K-$15K → $8,000, etc.). Sector tagging: 8-ticker-per-sector whitelist first; fall back to 25-keyword dictionary on asset name, most-matched-keywords wins. 419 of 2,284 trades (18%) receive a tag.

03 BUILD SIGNAL

Weekly heat score

Aggregate tagged trades to weekly counts per sector: $x(s, t)$ = # trades in sector s during week t. Rolling z-score with a 26-week trailing window: $heat(s, t) = (x(s, t) - \mu)/\sigma$. Lag by one week so the week-t signal never uses within-week or later information. Critical for avoiding look-ahead bias.

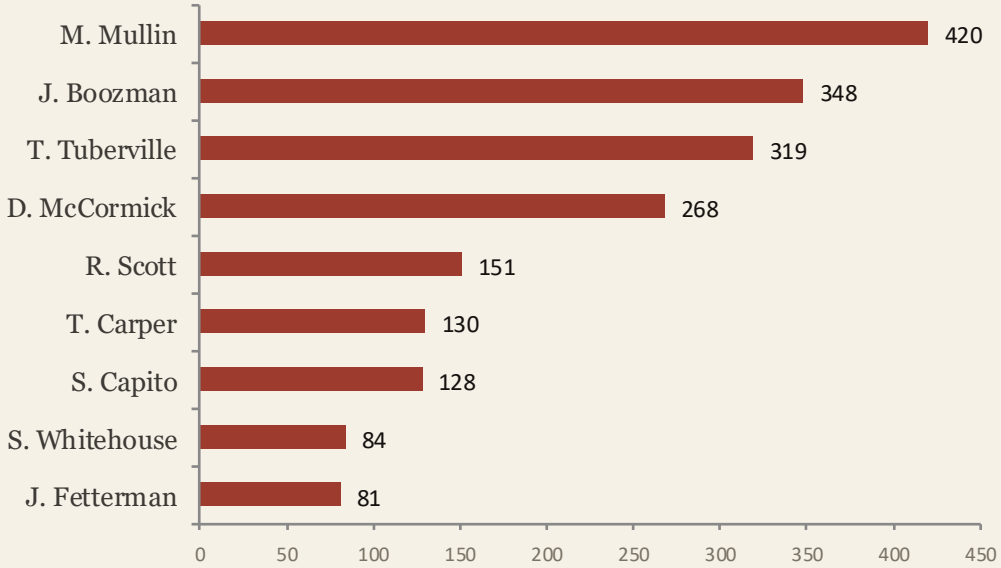
04 TEST

Statistical inference

For each sector s and horizon $h \in \{1, 5, 20\}$ trading days, OLS regression of forward return on lagged heat: $r(t, t+h) = \alpha + \beta \cdot heat(s, t-1) + \varepsilon$. Newey-West (HAC) standard errors with max(1.5h, 5) lags correct for serial correlation in weekly signals. 20 primary specifications + 10 sensitivity tests (dollar-weighted signal, net-buy count, out-of-sample split).

Three senators drove half the activity.

The aggregate masks what is really a small number of active traders.



Top 9 senators by trade count

47.6%

of all trades, three senators.

2,284

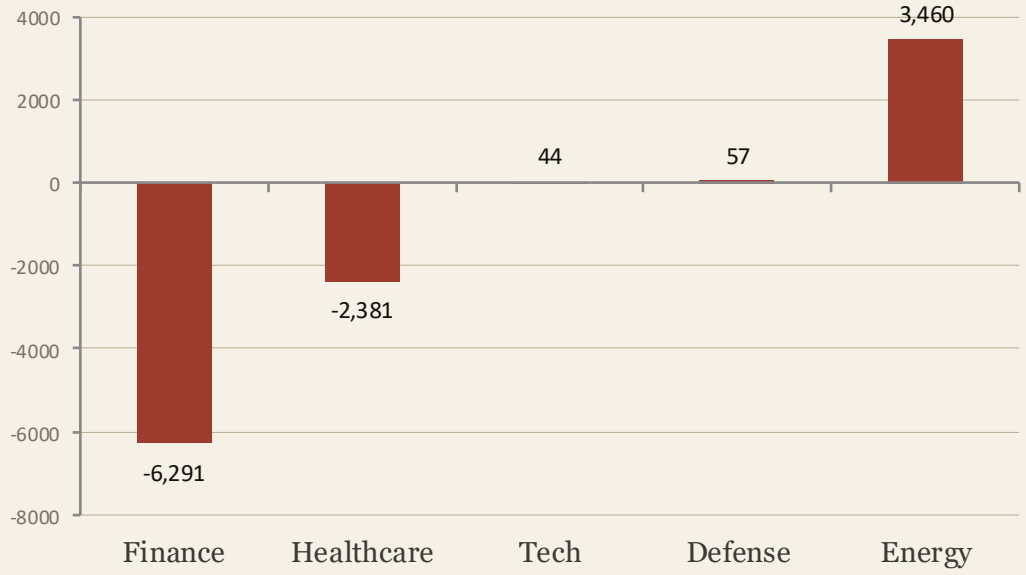
disclosed Senate trades, 2023–2026.

$207M

total notional (bucket midpoints).

Senators net-sold healthcare and finance.

Dollar-weighted net flow (buys minus sells) across 419 sector-tagged trades.



Net $ flow by sector, $ thousands

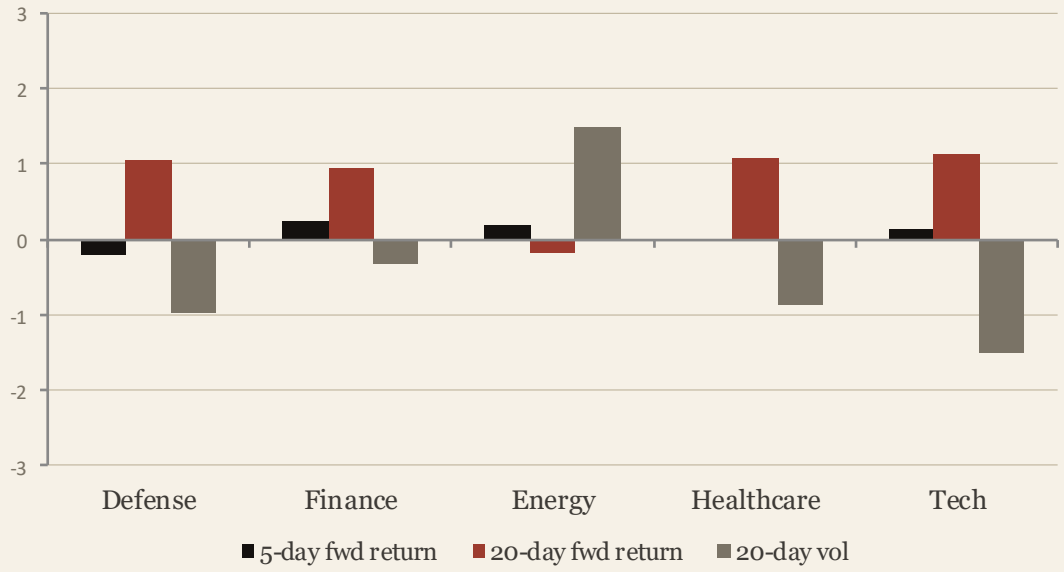
BUY SHARE BY SECTOR

% of trades that were purchases

Healthcare	***32%***	*24 buy / 50 sell*
Tech	***44%***	*78 / 98*
Finance	***51%***	*39 / 38*
Energy	***53%***	*43 / 38*
Defense	***56%***	*5 / 4*

No persistent predictive signal.

20 regressions: five sectors, four outcome windows each. The trade signal was lagged by one week, so it never peeked at the future



The |t|=1.96 significance threshold would sit just above 2 on the y-axis, none of the bars reach it.

0 / 20

tests significant at $p < 0.05$.

A secondary specification (raw net-buy count, no z-score) produced one hit, healthcare 20-day, $p = 0.03$. It did not replicate in an out-of-sample split (2nd half $p = 0.46$). Driven by one or two large sales.

The edge has faded.

The null hypothesis is consistent with the post-STOCK-Act literature. The 2024 headlines are driven by outliers, and by holding over trading.

1993–2008

PRE-STOCK ACT

+12%

per year

Ziobrowski et al. (2004, 2011). Senate portfolios beat the market by ~12% annually; House by ~6%. No routine disclosure, no 45-day clock.

2012–2020

POST-STOCK ACT

~0

abnormal return

Belmont, Sacerdote et al. (NBER 2020). Senate purchases 2012–2020 slightly underperformed. No evidence of stock-picking skill tied to committees.

2024–2025

RECENT HEADLINES

32.2%

of members beat SPY in 2025

Unusual Whales 2024/2025 reports. 2024 saw many beat the market, but the top performer (+104%) held NVDA-heavy ETFs, not active trading.

The takeaway: *post-disclosure, the active-trading edge largely disappears. Our 2023–2026 null fits the pattern.*

What made this hard.

I Senate EFD scraping

CSRF + cookie handshake required. Terms-of-use POST before any data request. Datacenter IPs blocked (Colab failed; local Mac worked). Older PTRs arrive as scanned PDFs that yield zero parseable rows, 20–40% empty rate.

II Congress.gov coverage mismatch

API defaults returned 5,000 bills but 99% were 2020–2021 or older housekeeping resolutions. Only 276 (5.5%) tagged to a sector, and almost none overlapped with the 2023–2026 trade window. The legislative half of the heat score was unusable.

III Small, concentrated sample

34 senators report; 3 account for 48% of activity. Defense sector: 9 total trades across three years. HAC standard errors inflate further with weekly autocorrelation. Power to detect a true effect is low. Itself a finding, not a bug.

IV Multiple-comparison discipline

20 primary tests + 10 sensitivity tests = 30 specifications at $\alpha = 0.05$. Expected Type-I errors by chance: ~1.5. Observed: 1. Resisted treating the healthcare hit as a finding before running the out-of-sample split that broke it.

The Data Science Process.

I started this project expecting to find an edge in Senate disclosures. That expectation shaped which specifications I tried first and which results I wanted to extend.

WHERE THE BIAS SHOWED UP

Following promising results too early.

When one specification produced a hit, I was inclined to build a narrative around it before confirming the result held under other reasonable choices.

WHAT THE PROCESS ADDED

Writing expectations down before running tests.

Stating the null up front and committing to out-of-sample checks made it easier to interpret results on their own terms rather than on whether they confirmed the prior.

Takeaway: *the methodology carries more weight than any single result. The null finding here is informative because the process was designed to let it be.*